

Lab 3 - Configuring a Multi-protocol network (OSPF, EIGRP, eBGP)

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Purpose -

Configuring BGP is an important concept to understand and is very important and frequently used knowledge for a future networking professional. This lab would allow students to configure various routing protocols as a review while requiring them to research about BGP to connect each different Autonomous System (AS).

Background Information -

This lab contains the usage of three different routing protocols. OSPF, EIGRP, and BGP.

OSPF -

OSPF has been used since 1991, meaning that it has been used for around 35 years now, and it has been widely used and adopted by many enterprises, data centers, and service providers.

A link-state routing protocol that uses a mathematical algorithm to determine the best path to every IP destination in the network. For example, if you have a network with a hundred routers running OSPF, each router knows how to reach every destination in the network via the most optimal path because using OSPF messages, each router advertises practically every detail about every connected link. This results in all routers collectively accumulating information about every device and network in the environment. Subsequently, every router knows how to reach every IP destination via the optimal network path.

OSPF separates the network into different areas, allowing different areas of the network to be controlled for different purposes and at different times. The backbone of this is 0.0.0.0, or Area 0. All other Areas must have some sort of connection to Area 0, or an Area Border Router (ABR), which retains different link-state databases for each area it borders.

Within this lab, OSPF was used twice in two different AS(es).

EIGRP –

Enhanced Interior Gateway Routing Protocol (EIGRP) is a Cisco-developed hybrid routing protocol that combines features of both distance vector and link-state protocols. It provides fast convergence, efficient routing updates, and advanced loop-prevention mechanisms. EIGRP employs a composite metric based on bandwidth and delay by default, with the option to include reliability and load.

EIGRP uses the Diffusing Update Algorithm (DUAL) to maintain a topology table of backup routes, enabling rapid failover in case of link failures. Instead of periodic updates, it sends incremental updates only when changes occur, reducing bandwidth usage. It uses multicast (224.0.0.10) for neighbor discovery and authentication to secure routing exchanges. Features such as automatic summarization, route filtering, and unequal-cost load balancing (variance) make it highly scalable and efficient for enterprise networks.

In this lab, configuring EIGRP involves setting up autonomous system numbers, verifying neighbor relationships and adjusting metric parameters. EIGRP remains a widely used IGP within Cisco environments due to its reliability and flexibility.

BGP –

External Border Gateway Protocol (eBGP) is an inter-domain routing protocol used to exchange routing information between autonomous systems (AS). It is a core component of internet routing, ensuring efficient communication between different organizations and service providers. Unlike interior gateway protocols (IGPs) like OSPF or EIGRP, eBGP makes routing decisions based on path attributes such as AS Path, Next-Hop, and Local Preference. It prevents routing loops by appending AS numbers to the AS Path attribute and filtering routes accordingly.

eBGP operates over TCP port 179 and requires manually configured neighbor relationships. By default, it assumes peers are directly connected and uses a Time-to-Live (TTL) of 1, but multi-hop BGP can be used for indirect connections. Route filtering, prefix lists, and route maps are commonly used

to enforce traffic engineering and security policies, however will not be used in this lab as the different AS(es) are direction Connected.

In this lab, configuring eBGP involves establishing peer relationships. Understanding how BGP propagates and manages routes is key.

Redistribution of Routes –

Route redistribution allows different routing protocols to share route information, enabling communication between networks running eBGP, EIGRP, and OSPF. Since these protocols use different metrics, redistribution requires defining appropriate values to ensure compatibility. Without proper filtering, it can lead to routing loops, excessive route propagation, or loss of connectivity. In this lab, students would apply redistribution commands and verifying connectivity to ensure connectivity.

Lab Summary –

The majority of this lab was setting up the different routing protocols, but the most difficult part was the troubleshooting. A major issue was the redistribution commands, as they were required everywhere for the network to distribute routes across the different AS(es). The base routing of each AS wasn't the difficult part, it was connecting them, as expected, because eBGP was a new concept to the students, and they had to mainly research what kind of commands they needed to configure on each router.

Lab Commands –

router bgp [autonomous-system-number]

- Creates and enters configuration mode for a BGP instance with the selected AS number.

neighbor [ip-address] remote-as [autonomous-system-number]

- Establishes a BGP neighbor. For iBGP, the remote autonomous system number must match the local AS.

Neighbor [ip-address] activate

network [*ip-address*] **mask** [*subnet-mask*]

- Advertises a specific network to other BGP peers. The network must be in the routing table to be advertised.

ip address [*ip-address*] [*subnet-mask*]

- Assigns an IP address to an interface.

address-family ipv4 [mdt | multicast | tunnel | unicast [*vrf vrf-name*]
| *vrf vrf-name*]

- Enters an address family configuration mode to configure a routing session using ipv4

address-family ipv6 [unicast | multicast | vpnv6] [*vrf vrf-name*]

- Enters an address family configuration mode to configure a routing session using ipv6

router ospf [*process-id*]

- Enables OSPF with a set process id

network [*ip-address*] [*wildcard-mask*] **area** [*area-id*]

- Defines OSPF networks and assigns them to a specific area.

ip route [*destination-ip*] [*subnet-mask*] [*next-hop-ip*]

- Creates a static route to ensure the reachability of BGP peer loopback addresses (if not using OSPF or EIGRP).

show ip bgp summary

- Displays a summary of BGP peers, including state and established sessions.

show ip bgp

- Shows the BGP routing table with learned and advertised routes.

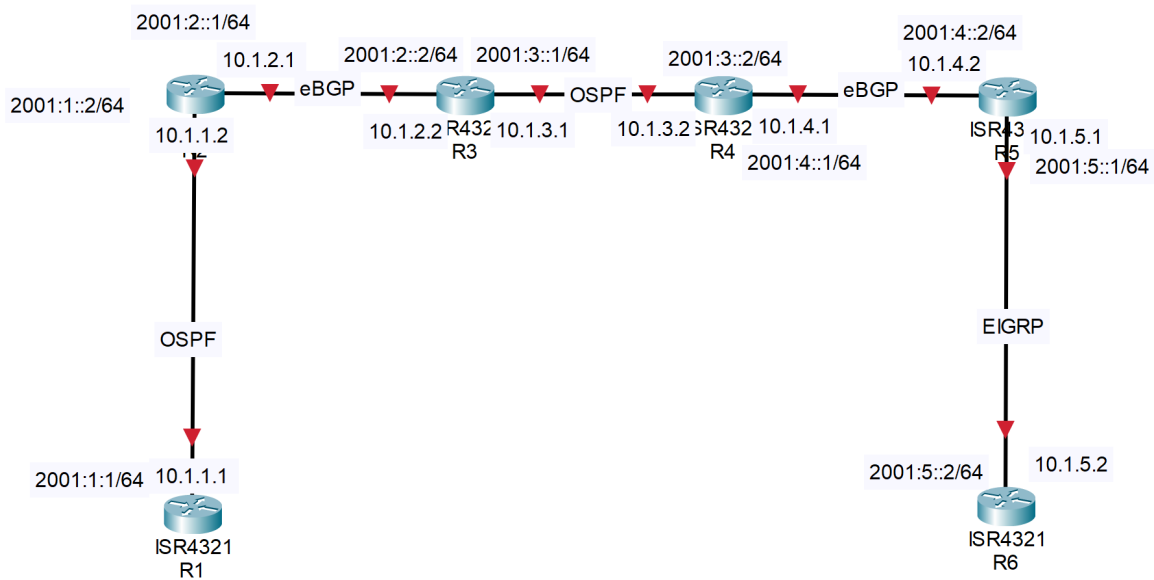
ping [*ip-address*] **source loopback0**

- Verifies reachability of the neighbor's loopback address using the local loopback as the source.

redistribute [*protocol*] [*autonomous-system-number*] [**metric** {*metric value*}] [*level*]

- Redistributes IPv6 routes from a specified protocol and gives it the specified metric.

Network Diagram -



Lab Configurations -

R1 -

Building configuration...

Current configuration : 3875 bytes

! Last configuration change at 01:07:19 UTC Wed Jan 22 2025

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-414644419
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-414644419
 revocation-check none
 rsakeypair TP-self-signed-414644419
license udi pid ISR4321/K9 sn FDO21080CTW
no license smart enable

```
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 10.1.1.1 255.255.255.0
negotiation auto
ipv6 address 2001:1::1/64
ipv6 ospf 1 area 0
interface GigabitEthernet0/0/1
ip address 10.1.0.1 255.255.255.0
negotiation auto
ipv6 address 2001::1/64
ipv6 ospf 1 area 0
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface Service-Engine0/2/0
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
```



```
router-id 1.1.1.1
redistribute connected subnets
network 10.1.1.0 0.0.0.255 area 0
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
  redistribute connected
control-plane
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
end
```

R2 -

Building configuration...

Current configuration : 2251 bytes

! Last configuration change at 00:57:07 UTC Wed Jan 22 2025

version 15.5

service timestamps debug datetime msec

```
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
    exit-address-family
  address-family ipv6
    exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214328EH
spanning-tree extend system-id
redundancy
  mode none
vlan internal allocation policy ascending
vlan 10,20
interface GigabitEthernet0/0/0
  ip address 10.1.1.2 255.255.255.0
  negotiation auto
  ipv6 address 2001:1::2/64
  ipv6 ospf 1 area 0
```

```
interface GigabitEthernet0/0/1
  ip address 10.1.2.1 255.255.255.0
  negotiation auto
  ipv6 address 2001:2::1/64
interface Serial0/1/0
  no ip address
  shutdown
interface Serial0/1/1
  no ip address
  shutdown
interface Service-Engine0/2/0
  no ip address
  shutdown
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
interface Vlan1
  no ip address
  shutdown
router ospf 1
  router-id 2.2.2.2
  redistribute connected subnets
  redistribute bgp 1 subnets
  network 10.1.1.0 0.0.0.255 area 0
  network 10.1.2.0 0.0.0.255 area 0
```

```
router bgp 1
  bgp log-neighbor-changes
  neighbor 10.1.2.2 remote-as 2
  neighbor 2001:2::2 remote-as 2
  address-family ipv4
    network 10.1.2.0
    redistribute connected
    redistribute ospf 1 match internal external 2
    neighbor 10.1.2.2 activate
    no neighbor 2001:2::2 activate
  exit-address-family
  address-family ipv6
    redistribute connected
    redistribute ospf 1 metric 10 match internal external 1 external 2
    network 2001:2::/64
    neighbor 2001:2::2 activate
  exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
  router-id 2.2.2.2
  redistribute connected
  redistribute bgp 1
control-plane
line con 0
```

```
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

R3 -

Building configuration...

Current configuration : 2322 bytes

! Last configuration change at 18:39:29 UTC Wed Jan 22 2025

version 15.5

service timestamps debug datetime msec

service timestamps log datetime msec

no platform punt-keepalive disable-kernel-core

hostname R3

boot-start-marker

boot-end-marker

vrf definition Mgmt-intf

address-family ipv4

exit-address-family

address-family ipv6

exit-address-family

no logging console

no aaa new-model

no ip domain lookup

ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214414TX
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
ip address 10.1.3.1 255.255.255.0
negotiation auto
ipv6 address 2001:3::1/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 10.1.2.2 255.255.255.0
negotiation auto
ipv6 address 2001:2::2/64
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto

```
interface GigabitEthernet0/2/1
  no ip address
  shutdown
  negotiation auto
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
interface Vlan1
  no ip address
  shutdown
router ospf 1
  router-id 3.3.3.3
  redistribute connected subnets
  redistribute bgp 2 subnets
  network 10.1.2.0 0.0.0.255 area 1
  network 10.1.3.0 0.0.0.255 area 1
router bgp 2
  bgp log-neighbor-changes
  neighbor 10.1.2.1 remote-as 1
  neighbor 2001:2::1 remote-as 1
  address-family ipv4
    network 10.1.2.0
    redistribute connected
    redistribute ospf 1 match internal external 1 external 2
    neighbor 10.1.2.1 activate
```

```
no neighbor 2001:2::1 activate
exit-address-family
address-family ipv6
redistribute connected
redistribute ospf 1 metric 10 match internal external 1 external 2
network 2001:2::/64
neighbor 2001:2::1 activate
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
router-id 3.3.3.3
redistribute connected
redistribute bgp 2
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

R4 –

Building configuration...

Current configuration : 4574 bytes

! Last configuration change at 13:14:34 UTC Wed Jan 22 2025

version 16.9

service timestamps debug datetime msec

service timestamps log datetime msec

platform qfp utilization monitor load 80

platform punt-keepalive disable-kernel-core

hostname R4

boot-start-marker

boot-end-marker

vrf definition Mgmt-intf

address-family ipv4

exit-address-family

address-family ipv6

exit-address-family

no aaa new-model

login on-success log

subscriber templating

ipv6 unicast-routing

multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-1275915594

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-1275915594

revocation-check none

rsakeypair TP-self-signed-1275915594

license udi pid ISR4321/K9 sn FDO214811ZM

```
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 10.1.3.2 255.255.255.0
negotiation auto
ipv6 address 2001:3::2/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 10.1.4.1 255.255.255.0
negotiation auto
ipv6 address 2001:4::1/64
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
```

```
negotiation auto
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
router ospf 1
  router-id 4.4.4.4
  redistribute connected subnets
  redistribute bgp 2 subnets
  network 10.1.3.0 0.0.0.255 area 1
  network 10.1.4.0 0.0.0.255 area 1
router bgp 2
  bgp log-neighbor-changes
  neighbor 10.1.4.2 remote-as 3
  neighbor 2001:4::2 remote-as 3
  address-family ipv4
    network 10.1.4.0
    redistribute connected
    redistribute ospf 1 match internal external 1 external 2
    neighbor 10.1.4.2 activate
    no neighbor 2001:4::2 activate
  exit-address-family
  address-family ipv6
    redistribute connected
    redistribute ospf 1 match internal external 1 external 2
    network 2001:4::/64
```

```
neighbor 2001:4::2 activate
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
router-id 4.4.4.4
redistribute connected
redistribute bgp 2
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

R5 –

Building configuration...

Current configuration : 4354 bytes

! Last configuration change at 01:10:23 UTC Wed Jan 22 2025

version 16.9

service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core
hostname R5
boot-start-marker
boot system flash bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker
vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2219300048
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-2219300048
 revocation-check none
 rsakeypair TP-self-signed-2219300048
license udi pid ISR4321/K9 sn FLM240607T3
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id

```
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 10.1.5.1 255.255.255.0
negotiation auto
ipv6 address 2001:5::1/64
ipv6 enable
ipv6 eigrp 3
interface GigabitEthernet0/0/1
ip address 10.1.4.2 255.255.255.0
negotiation auto
ipv6 address 2001:4::2/64
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router eigrp 3
default-metric 10000 100 255 1 1500
network 10.1.5.0 0.0.0.255
redistribute connected
redistribute bgp 3
eigrp router-id 5.5.5.5
router bgp 3
bgp log-neighbor-changes
neighbor 10.1.4.1 remote-as 2
neighbor 2001:4::1 remote-as 2
```

```
address-family ipv4
network 10.1.4.0
redistribute connected
redistribute eigrp 3
neighbor 10.1.4.1 activate
no neighbor 2001:4::1 activate
exit-address-family
address-family ipv6
redistribute connected
redistribute eigrp 3
network 2001:4::/64
neighbor 2001:4::1 activate
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 3
eigrp router-id 5.5.5.5
redistribute connected
redistribute bgp 3
default-metric 10000 100 255 1 1500
control-plane
line con 0
transport input none
stopbits 1
```

```
line aux 0
  stopbits 1
line vty 0 4
  login
end
```

R6 SHOW RUNNING CONFIGURATIONS:

Building configuration...

Current configuration : 1831 bytes

! Last configuration change at 00:42:16 UTC Wed Jan 22 2025

version 15.5

service timestamps debug datetime msec

service timestamps log datetime msec

no platform punt-keepalive disable-kernel-core

hostname R6

boot-start-marker

boot-end-marker

vrf definition Mgmt-intf

address-family ipv4

exit-address-family

address-family ipv6

exit-address-family

no aaa new-model

ipv6 unicast-routing

subscriber templating

vtp domain cisco

vtp mode transparent


```
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214913GF
spanning-tree extend system-id
redundancy
  mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
  ip address 10.1.5.2 255.255.255.0
  negotiation auto
  ipv6 address 2001:5::2/64
  ipv6 enable
  ipv6 eigrp 3
interface GigabitEthernet0/0/1
  ip address 10.0.5.1 255.255.255.0
  negotiation auto
  ipv6 address 2001:5:1::1/64
  ipv6 eigrp 3
interface Serial0/1/0
  no ip address
  shutdown
interface Serial0/1/1
  no ip address
  shutdown
interface GigabitEthernet0/2/0
  no ip address
  shutdown
  negotiation auto
```

```
interface GigabitEthernet0/2/1
  no ip address
  shutdown
  negotiation auto
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
interface Vlan1
  no ip address
  shutdown
router eigrp 3
  default-metric 10000 100 255 1 1500
  network 10.1.5.0 0.0.0.255
  redistribute connected
  eigrp router-id 6.6.6.6
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 3
  eigrp router-id 6.6.6.6
  redistribute connected
  default-metric 10000 100 255 1 1500
control-plane
line con 0
```

```
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Issues –

There was an issue where the OSPF routes were EX2 OSPF routes, which students had to configure the redistribution command of OSPF from “redistribute OSPF 1” to “redistribute ospf 1 match internal 1 external 1 external 2” in order to distribute all routes. The initial setup of each address family was also quite confusing for the students, as it was again a new concept.

Conclusion –

This lab provided a great review on old IGPs, OSPF and EIGRP. The students configured six routers, four had OSPF, four also had BGP, and two had EIGRP. This lab gives them an introduction to BGP, a popular routing protocol that is frequently used by companies and gives them an experience that is more similar to a real scenario. As the students progressed through the lab, it also seems as though they were becoming more and more confident in their network creation, subnetting, and the configuration of routing protocols.

Lab Sign Off –

01/21/2025

Configuring a Network with OSPF, EIGRP, and eBGP

Andy Yang

P3-4 CCNP

Mr. Mason

